

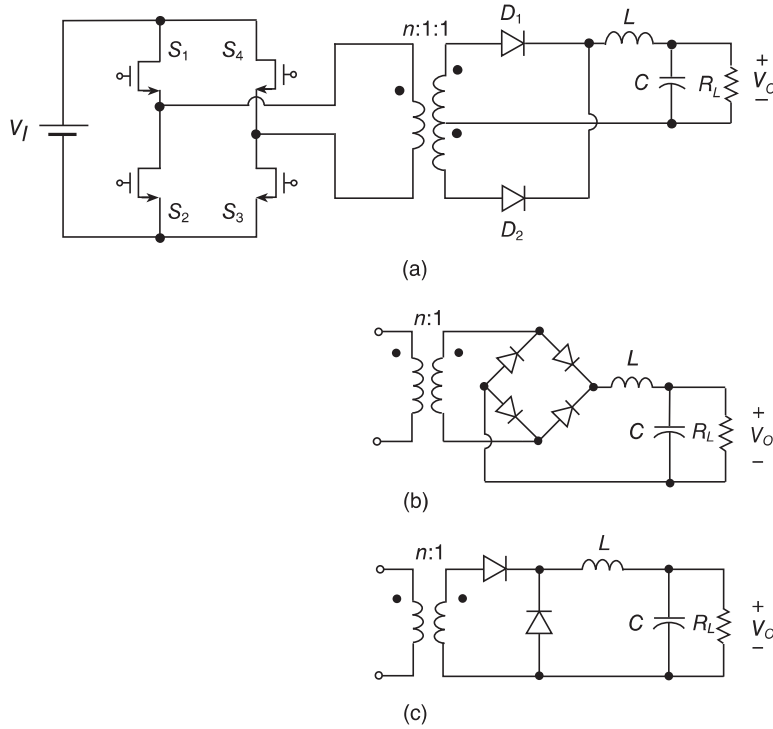


Figure 8.1 Full-bridge converter. (a) With a transformer center-tapped rectifier. (b) With a full-bridge rectifier. (c) With a half-wave rectifier.

large enough to reduce the current through this inductance. On the other hand, if the magnetizing inductance is too large, it requires a large number of turns and is physically large. Ideally, the dc component of the current through the magnetizing inductance is zero. A coupling capacitor may be added in series with the primary winding to achieve zero dc component of the current through the magnetizing inductance and thereby removing an imbalance of the magnetic core. The full-bridge converter is well suited for high-power applications, usually from 0.5 kW to several kilowatts. It offers the highest power levels among all converters. In very high-power applications, insulated-gate bipolar transistors (IGBTs), thyristors, or MOSFET-controlled thyristors (MCTs) are used as switching devices. In addition, two or more power switching devices may be connected in parallel to increase current capability of every switch and output power levels.

Three topologies can be used for the full-bridge converter: with a transformer center-tapped rectifier, with a full-bridge rectifier, or with a half-wave rectifier. The transformer center-tapped rectifier consists of two diodes D_1 and D_2 , an inductor L , a filter capacitor C , and a load resistor R_L . IGBTs or MCTs can also be used. This rectifier is most suitable for low output voltage applications because only one diode conducts, when two switches are ON. Schottky diodes or low on-resistance power MOSFETs can be used as rectifying devices. The voltage stress of the diodes is $2V_I/n$, which is higher than that in the bridge rectifier. Therefore, the transformer center-tapped rectifier is not suitable for high-voltage applications.

The bridge rectifier is suitable for high output voltage applications because the voltage stress of the diodes is V_I/n , which is half of the transformer center-tapped rectifier. This rectifier is not suitable for low-voltage applications because the two diodes conduct when the two switches are ON, and the total forward voltage across the two diodes may become comparable with the output voltage, resulting in low efficiency.